FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – V
PAPER –2
TEST DATE: 08-02-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

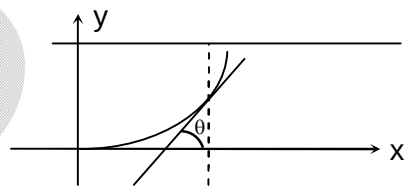
1. D

Sol. Equation of circle is

$$(x-0)^2+(y-R)^2=R^2 \quad \ldots(1)$$

$$\frac{dy}{dx}=\tan\theta \quad \ldots(2)$$

By Snell's law $\sin(90-\theta)\times\eta=1\times\eta_0$ …(3)



2. D

Sol. $F=-\left(\frac{Ay}{L}\right)\Delta L=K\Delta L$

$\Rightarrow \quad K=\frac{Ay}{L}$

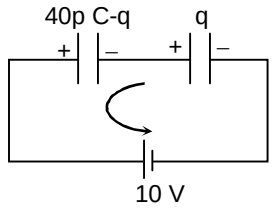
$T=2\pi\sqrt{\frac{m}{K}}$

3. A

Sol. Apply KVL

$$\frac{40-q}{2}-10-\frac{q}{2}=0$$

Solve for q



4. B

Sol. Velocity, $v=\sqrt{\frac{YRT}{M}}$

Find Y

5. B, C, D

Sol. $U = U_0(1-\cos ax)$, $F = \dfrac{-dU}{dx} = -aU_0 \sin ax$

$m\dfrac{d^2x}{dt^2} = -a^2U_0x$ (For small oscillations)

$\Rightarrow T = 2\pi\sqrt{\dfrac{m}{a^2U_0}}$

Potential energy is maximum at $x = \dfrac{\pi}{a}$

amplitude is $\dfrac{\pi}{a}$

Speed of particle is maximum at $x = 0$ ($\because$ PE is minimum)

6. A, C

Sol. (i) for $\theta = 30$

$\lambda = \dfrac{c}{\nu} = \dfrac{3\times10^8}{10^6} = 300$ m, d = 150 m

$\therefore \quad \delta = \dfrac{2\pi}{\lambda}(d\sin\theta) = \dfrac{2\pi}{(300)}(150)\left(\dfrac{1}{2}\right) = \dfrac{\pi}{2}$

$I(\theta) = I_0\cos^2\left(\dfrac{\delta}{2}\right) = I_0\cos^2\left(\dfrac{\pi}{4}\right) = \dfrac{I_0}{2}$

(ii) for $\theta = 90° \Rightarrow \delta = \left(\dfrac{2\pi}{300}\right)(150)(1) = \pi$

$\therefore \quad I(\theta) = I_0\cos^2\left(\dfrac{\delta}{2}\right)$

$= I_0\cos^2\left(\dfrac{\pi}{2}\right) = 0$

(iii) for $\theta = 0°$, $\delta = 0$

$\therefore \quad I(\theta) = I_0$

7. A, D

Sol. Change in angular momentum = angular impulse

$L = 5\tau$

$I = \dfrac{ma^2}{3}$

$K = \dfrac{L^2}{2I} = \dfrac{75\tau^2}{2ma^2}$

8. A, C

Sol. $a = \dfrac{F_0}{m}\cos\omega t$, $\displaystyle\int_0^0 dv = \frac{F_0}{m}\int_0^{t_0} \cos\omega t\ dt$, $t_0 = \dfrac{\pi}{\omega}$

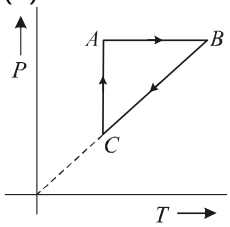
and $\displaystyle\int_0^v dv = \frac{F_0}{m}\int_0^t \cos\omega t\ dt$, $v = \dfrac{F_0}{m\omega}\sin\omega t$, $\displaystyle\int_0^s ds = \frac{F_0}{m\omega}\int_0^{t_0} \sin\omega t\, dt$

$S = \dfrac{F_0}{m\omega^2}\left[-\cos\omega t\right]_0^{t_0} = \dfrac{F_0}{m\omega^2}[2] = \dfrac{2F_0}{m\omega^2}$

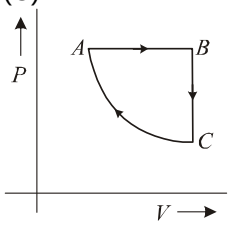
9. A, C

Sol. Using PV = nRT

(A)



(C)



10. A, B, C, D

Sol. Charge on capacitor at time t is

$q = Q_0 \cos \omega t$

Where $\omega = \frac{1}{\sqrt{LC}}$.

At $\omega t = \frac{\pi}{2}, q = 0$

$\Rightarrow \quad t = \frac{\pi}{2}\sqrt{LC}$

By conservation of energy

$\frac{Q_0^2}{2C} = \frac{1}{2}Li_{max}^2$

$\Rightarrow \quad i_{max} = \frac{Q_0}{\sqrt{LC}}$

Section – B

11. 5

Sol. $(\Delta Q)_P = nC_P\Delta T$

When piston is clamped; heat is rejected and volume becomes constant

$(\Delta Q)_V = nC_V\Delta T$

Difference = $nC_P\Delta T - nC_V\Delta T = \Delta W = P\Delta V$

$P = P_0 + \frac{Mg}{A}$ and $\Delta V = (A \times 0.1)$

$\Rightarrow (\Delta Q)_P - (\Delta Q)_V = \left(P_0 + \frac{Mg}{A}\right) \times A \times 0.1$

= 55 J

$\Rightarrow x = 5$

12. 4

Sol. $\sqrt{2}\ \lambda_0 T_1 = \lambda_0 T_2$

$$\frac{T_2}{T_1} = \sqrt{2} \quad ; \quad \frac{A_2}{A_1} = \left(\frac{T_2}{T_1}\right)^4 = 4$$

13. 1

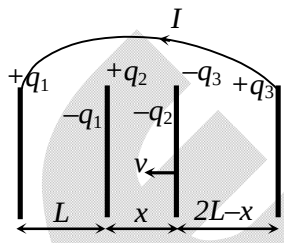
Sol. $$\frac{2q_1 L}{2A\varepsilon_0} + \frac{2q_2 x}{2A\varepsilon_0} - \frac{2q_3(2L-x)}{2A\varepsilon_0} = 0 \qquad \ldots \text{(i)}$$

$$I = \frac{dq_1}{dt} = \frac{-dq_3}{dt} = \frac{dq_2}{dt} \qquad \ldots \text{(ii)}$$

From (i), $q_1 L + q_2 x - 2q_3 L + q_3 x = 0$

$$\Rightarrow \quad L\frac{dq_1}{dt} + x\frac{dq_2}{dt} + q_2\frac{dx}{dt} - 2\frac{dq_3}{dt}L + x\frac{dq_3}{dt} + q_3\frac{dx}{dt} = 0$$

$$\Rightarrow \quad I = \frac{v}{3L}(q_2 + q_3) = 1\ \text{Ma}$$



14. 2

Sol. Apply Bernoulli's equations at pipe (1), (2) and (3)

$$p_3 + \frac{1}{2}\rho v_3^2 = p_2 + \frac{1}{2}\rho v_2^2 = p_1 + \rho/2 v_1^2$$

$$p_0 + \rho g h_3 + \frac{1}{2}\rho v_3^2 = p_0 + \rho g h_2 + \frac{\rho}{2}v_2^2 = p_0 + \rho g h_1 + \frac{\rho}{2}v_1^2$$

$$g h_3 + \frac{v_3^2}{2} = g h_2 + \frac{v_2^2}{2} = g h_1 + \frac{v_1^2}{2} \qquad \ldots \text{(i)}$$

$$v_3 A_3 = v_2 A_2$$

$$v_3 \times A = v_2 \times 2 \qquad \ldots \text{(ii)}$$

$$v_2 \times 2 = v_1 \times 1 \qquad \ldots \text{(iii)}$$

From Equation (i), we get

$$g h_2 + \frac{v_2^2}{2} = g h_1 + \frac{v_1^2}{2}$$

$$10 \times \frac{35}{100} + \frac{v_2^2}{2} = 10 \times \frac{20}{100} + \frac{v_1^2}{2}$$

$$\frac{15}{10} = \frac{v_1^2 - v_2^2}{2}$$

$$= \frac{4v_2^2 - v_2^2}{2}$$

$$= \frac{3v_2^2}{2}$$

$v_2 = 1$ m/s, $v_1 = 2$ m/s

From equation (i), we get

$$g h_3 + \frac{v_3^2}{2} = g h_2 + \frac{v_2^2}{2}$$

$$= 10 \times \frac{37.5}{100} + \frac{v_3^2}{2}$$

$$= 10 \times \frac{35}{100} + \frac{1}{2}$$

$$\frac{v_3^2}{2}=\frac{1}{2}+\frac{35}{10}-\frac{37.5}{10}=\frac{1}{2}-\frac{2.5}{10}=\frac{1}{2}-\frac{25}{100}$$

$$\frac{v_3^2}{2}=\frac{1}{4}\Rightarrow v_3^2=\frac{1}{2}$$

$$\Rightarrow \quad v_3=\frac{1}{\sqrt{2}}, X=2$$

15. 2

Sol.
$$\sqrt{v}=a(Z-b)$$

$$\sqrt{v}=a(Z-1)$$

$$\Rightarrow \quad (Z-1)^2\propto \nu$$

or
$$(Z-1)^2\propto\frac{1}{\lambda}$$

$$\Rightarrow \quad \frac{(Z_{Mo}-1)^2}{(Z_{Cu}-1)^2}=\frac{\lambda_{Cu}}{\lambda_{Mo}}$$

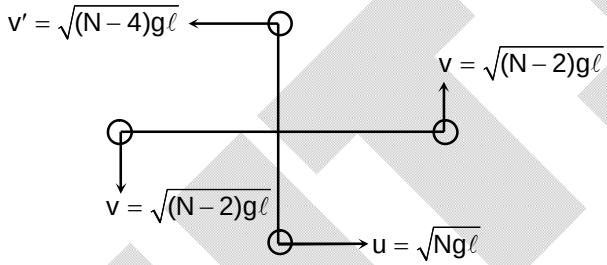
$$\Rightarrow \quad \lambda_{Cu}=\lambda_{Mo}\frac{(Z_{Mo}-1)^2}{(Z_{Cu}-1)^2}$$

$$=0.71\times\left(\frac{41}{28}\right)^2=1.52\text{Å}$$

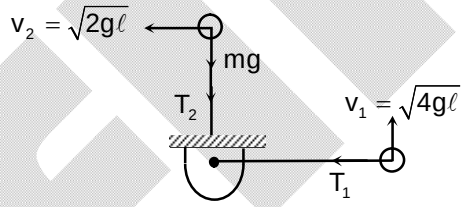
λ = 2 Å

16. 4

Sol. For vertical circular motion, student should always remember that



For given, $u=\sqrt{6g\ell}$



At horizontal position,

$$T_1=\frac{mv_1^2}{\ell}=4mg \qquad \ldots(i)$$

At highest point,

$$T_2+mg=\frac{mv_2^2}{\ell}=2\,mg$$

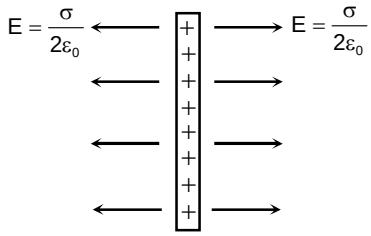
$$T_2=mg \qquad \ldots(ii)$$

From above equation, we get

$$\frac{T_1}{T_2}=4$$

17. 4

Sol. The electric field produced by thin charged sheet is given by $\frac{\sigma}{2\varepsilon_0}$.



[Where, σ = surface charge density]

$E_0 + E = 12$ V/m ...(i)

$E_0 - E = 8$ V/m ...(ii)

Solving Equations (i) and (ii), we get

$E = 2$ V/m

Now, electric field due to plate $= \frac{\sigma}{2E_0} = 2$

$s = 4e_0$

$\therefore$ n = 4

18. 6

Sol. From the given conditions, we get

$E_n - E_2 = (10.2 + 17)eV$

= 27.2 eV ...(i)

$E_n - E_3 = (4.25 + 5.95)eV$

= 10.2 eV ...(ii)

Subtracting Equation (ii) from Eq. (i), we get

$E_3 - E_2 = 17.0$ eV

$Z^2(13.6)\left[\frac{1}{4} - \frac{1}{9}\right] = 17.0$

$Z^2(13.6)\left(\frac{5}{36}\right) = 17.0$

$Z^2 = 9$ or Z = 3

$Z^2(13.6)\left[\frac{1}{4} - \frac{1}{n^2}\right] = 27.2$

$3^2(13.6)\left[\frac{1}{4} - \frac{1}{n^2}\right] = 27.2$

$\frac{1}{4} - \frac{1}{n^2} = 0.222$

$\Rightarrow \quad \frac{1}{n^2} = 0.0278 \Rightarrow n^2 = 36 \Rightarrow n = 6$

Chemistry

PART – II

Section – A

19. D
Sol. sp^2 hybridization of Hg and Hg^{2+} has d^{10} configuration

20. A
Sol. MnO_2 oxidise allylic alcohols

21. B
Sol. Factual

22. A
Sol. Factual

23. C, D
Sol. Factual

24. A, C, D
Sol. Down the group nucleophilicity increases

25. A, D
Sol. $-NO_2$ and –Br increases acidic character of phenol by dispersing –ve charge of phenoxide ion – OCH_3 and $-CH_3$ decreases acidic character by intensification of –ve charge of phenoxide ion.

26. A, D
Sol. Factual

27. A, C
Sol. p-subshell can have maximum 6 electron and an orbital can have maximum 2 electron with opposite spin.

28. A, C
Sol. Factual

Section – B

29. 7
Sol. for s-subshel, l = zero, hence for all electrons presente in s-orbitals, n × l = zero
Cr = $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $3d^5$, $4s^1$
In Cr there are 7 electrons present in s-orbitals

30. 6
Sol. Factual

31. 6
Sol. Triangular bipyramidal molecules have six, 90º angles

32. 5
Sol. edta make five rings of 5 member, x = 5 and octahydral [M(edta)] complex can not show geometrical isomerism, y = 0

33. 6

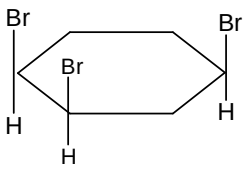
Sol. $Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \longrightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$

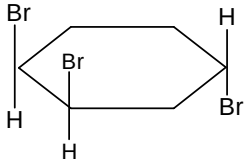
34. 4

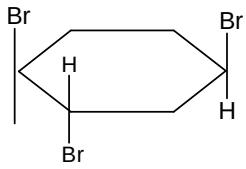
Sol. $Z = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$

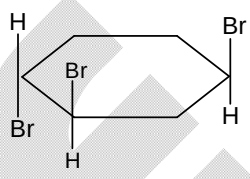
35. 4

Sol.



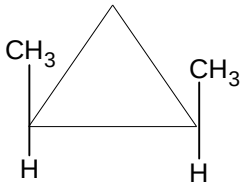


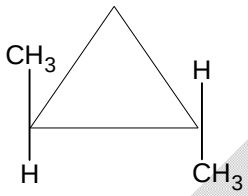


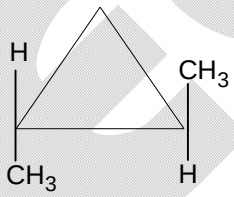


36. 3

Sol.







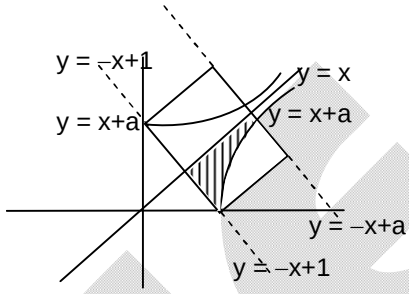
Mathematics

PART – III

Section – A

37. D

Sol. Required area of shaded region is $k-\frac{1}{4}$

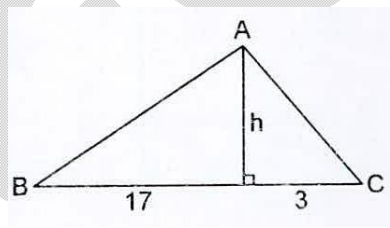


38. B

Sol. $B + C = \pi - A$

$\tan(B + C) = -\tan A$

$$\frac{h\left(\frac{1}{17}+\frac{1}{3}\right)}{1-\frac{h^2}{51}} = -\frac{22}{7} \Rightarrow \left(\frac{20h}{51-h^2}\right) = -\frac{22}{7}$$



Area of triangle ABC $\Delta = -\frac{1}{2}\times 20\times 11 = 110$

39. B

Sol. Let three digits of N be :

Hundred's digit = a – d, ten's digit = a, and unit's digit = a + d

$(a - d) + a + (a + d) = 15 \Rightarrow a = 5$

$N = 100(5 - d) + 50 + (5 + d)$ …(1)

Number obtained by reversing digits of N is

$N_1 = 100(5 + d) + 50 + (5 - d)$ …(2)

Now, $594 = N - N_1 = 10(-2d) + 2d$

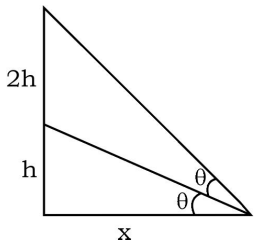
$\Rightarrow 2d = -6 \quad \Rightarrow d = -3$

Thus, N = 852 and $\frac{1000}{N-252} = \frac{1000}{600} = \frac{5}{3}$

40. C

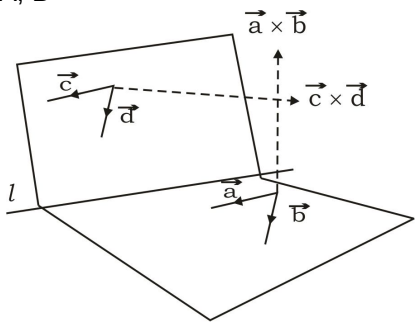
Sol. Let $\frac{h}{x} = a$. Then $\tan\theta = a, \tan 2\theta = 3a$.

$$\therefore \quad \frac{2a}{1-a^2} = 3a \quad \Rightarrow \quad a = \frac{1}{\sqrt{3}}$$



41. A, B

Sol.



$(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) =$ a vector perpendicular to $\vec{a}\times\vec{b}$ and $\vec{c}\times\vec{d}$

If l is the line of intersection of the two planes of $\vec{a},\vec{b}$ and $\vec{c},\vec{d}$ then $l \perp \vec{a}\times\vec{b}$ and $l \perp \vec{c}\times\vec{d}$. So, $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})$ is along l which is equally inclined to $\vec{a}\times\vec{b}$ and $\vec{c}\times\vec{d}$

42. A, B, C, D

Sol. (A) ${}^7C_2 - {}^3C_2 - {}^5C_2 + 2 = 10$

(B) ${}^7C_3 - {}^3C_3 - {}^5C_3 = 24$

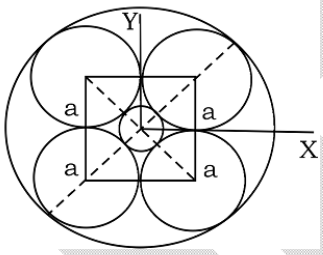
(C) There is no use of point'3'

${}^6C_4 - {}^4C_4 - {}^4C_3.{}^2C_2 = 6$

(D) ${}^7C_3 - {}^3C_3 - {}^5C_3 = 24$

43. A, C

Sol. The four circles are as shown in the figure. The smallest circle touching all of them has the radius = $\sqrt{2}a - a$ and the greatest circle touching all of them has the radius = $\sqrt{2}a + a$.



44. A, C

Sol. Consider $f(x) = x^3 + ax^2 + bx - 2$

$\alpha\beta\gamma = 2$

$\alpha, \beta, \gamma \in N \Rightarrow$ two of them are 1 & other is 2

$\Rightarrow a = -(\alpha+\beta+\gamma) = -4 = \dfrac{f''(0)}{2}$

$b = (\alpha\beta+\beta\gamma+\alpha\gamma) = 5 = f'(0)$

Hence $f(x) = x^3 - 4x^2 + 5x - 2$

Again $\because$ 1 is a repeated root of f(x) = 0

$\Rightarrow \quad f'(1) = 0$

45. A, B, C, D

Sol. $x + [y] + \{z\} = 1.1$ …(1)

$[x]+\{y\}+z=2.2$ ...(2)

$\{x\}+y+[z]=3.3$ ...(3)

(1) + (2) + (3)

$\Rightarrow \quad 2(x+y+z)=6.6$

$\Rightarrow \quad x+y+z=3.3$ ...(4)

(4) – (1)

$\{y\}+[z]=2.2 \quad \Rightarrow \{y\}=0.2 \,\&\, [z]=2$

(4) – (2)

$\{x\}+[y]=1.1 \quad \Rightarrow \{x\}=0.1,\ [y]=1$

(4) – (3)

$[x]+\{z\}=0 \quad \Rightarrow [x]=0 \ \& \ \{z\}=0$

$$\begin{cases} x=0.1 \\ y=1.2 \\ z=2 \end{cases}$$

46. B, D

Sol. $b^2=ac$

$$\cos B=\frac{a^2+c^2-b^2}{2ac}=\frac{a^2+c^2-b^2}{2b^2}$$

$\because \quad \cos B=\dfrac{a^2+c^2}{2ac}-\dfrac{1}{2}$

$$\Rightarrow \cos B+\frac{1}{2}=\frac{a^2+c^2}{2ac} \Rightarrow \cos B+\frac{1}{2}\geq 1 \Rightarrow \cos B\geq\frac{1}{2}$$

$B\in(0,\pi/3)$

Section – B

47. 9

Sol. $S=3.3^{2/2}.3^{3/4}.3^{4/8}.....\infty$

$S=3^{1+\frac{2}{2}+\frac{3}{4}+\frac{4}{8}+....\infty}$

$S'=1+\dfrac{2}{2}+\dfrac{3}{4}+\dfrac{4}{8}+....\infty$ ....(1)

$\dfrac{S'}{2}=\dfrac{1}{2}+\dfrac{2}{4}+\dfrac{3}{8}+....\infty$ ....(2)

Subtract (2) from (1)

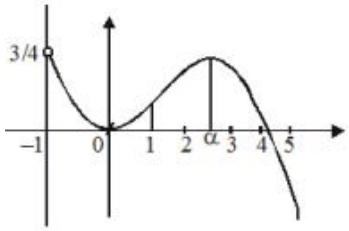
$\dfrac{S'}{2}=1+\dfrac{1}{2}+\dfrac{1}{4}+\dfrac{1}{8}+....\infty$

$S'=$

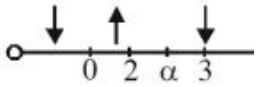
$S=3^4=81$

48. 2

Sol. $f(x)=(1+x)\ell n(1+x)-x-\dfrac{x^2}{4}$



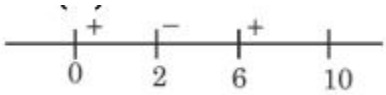
$\lim\limits_{x\to -1} f(x) = 0 + 1 - \frac{1}{4} = \frac{3}{4}$ and $f(0) = 0$



$\Rightarrow \alpha \in (2, 3)$

49. 2

Sol.



50. 9

Sol. Projection length = $|\vec{a}.\vec{p}|$

So, $\frac{|x + 2y + 3z|}{\sqrt{14}} = 6$

$\Rightarrow |x + 2y + 3z| = 6\sqrt{14}$

$\Rightarrow |(x\hat{i} + \sqrt{2}y\hat{j} + \sqrt{3}z\hat{k}).(\hat{i} + \sqrt{2}\hat{j} + \sqrt{3}\hat{k})| = 6\sqrt{14}$

$\Rightarrow (x^2 + 2y^2 + 3z^2)(1 + 2 + 3)\cos^2\theta = \left(6\sqrt{14}\right)^2$

$\Rightarrow \frac{x^2 + 2y^2 + 3z^2}{\sqrt{14}} \geq 6\sqrt{14} \Rightarrow I = 6\sqrt{14}$

So, $(I^2 + 15^2)^{1/3} = (504 + 225)^{1/3} = (729)^{1/3} = 9$.

51. 0

Sol. $\vec{c} = \frac{\vec{a} - \mu\vec{b}}{4}$ and $\vec{a}.\vec{b} = 0$

Now, $\left(\vec{b} - \vec{a}\right).\left(\vec{b} + \vec{c}\right) = 0 \Rightarrow \left(\vec{b} - \vec{a}\right).\left(\vec{b} + \frac{\vec{a} - \mu\vec{b}}{4}\right) = 0$

$\Rightarrow (4 - \mu)b^2 = a^2$ ($\because \mu < 4$) ... (i)

Again $4|\vec{b} + \vec{c}|^2 = |\vec{b} - \vec{a}|^2 \Rightarrow 4\left|\frac{(4-\mu)\vec{b} + \vec{a}}{4}\right|^2 = |\vec{b} - \vec{a}|^2$

$\Rightarrow 4\left(\frac{4-\mu}{4}\right)^2 b^2 + \frac{a^2}{4} = b^2 + a^2 \Rightarrow ((4 - \mu)^2 - 4)b^2 = 3a^2$ ... (ii)

(i) & (ii) we get $\frac{(4-\mu)^2 - 4}{4-\mu} = \frac{3}{1} \Rightarrow \mu^2 - 5\mu = 0$

$\Rightarrow \mu = 0$ or 5 but as $\mu < 4$, so, $\mu = 0$.

52. 8

Sol. $p = \dfrac{5!-{}^3C_1(4!).(2!)+{}^3C_2(3!).(2!).(2!)-{}^3C_3(2!).(2!).(2!).(2!)}{5!} = \dfrac{120-144+72-16}{120} = \dfrac{4}{15}$

So $30p = 8$

53. 2

Sol. $\int \underset{I}{x}\, f''\underset{II}{(2x)}dx = x\dfrac{f'(2x)}{2} - \int \dfrac{f'(2x)}{2}dx = x\dfrac{f'(2x)}{2} - \dfrac{f(2x)}{4}$

$$\int_0^1 x f''(2x)dx = \left|\frac{xf'(2x)}{2} - \frac{f(2x)}{4}\right|_0^1 = \left(\frac{f'(2)}{2} - \frac{f(2)}{4}\right) - \left(0 - \frac{f(0)}{4}\right)$$

$$\left(\frac{5}{2} - \frac{3}{4}\right) - \left(0 - \frac{1}{4}\right) = 2$$

54. 1

Sol. Let E_1, E_2, E_3 E_4, E_5 and E_6 be the events of occurrence of 1, 2, 3, 4, 5 and 6 on the dice respectively, and let E be the event

$$\therefore P(E_1) = \frac{1-k}{6};\ P(E_2) = \frac{1+2k}{6};\ P(E_3) = \frac{1-k}{6}$$

$$P(E_4) = \frac{1+k}{6};\ P(E_5) = \frac{1-2k}{6};\ P(E_6) = \frac{1+k}{6} \text{ and } \frac{1}{9} \le P(E) \le \frac{2}{9}$$

Then, $E \equiv \{(3,6), (6,3), (4,5), (5,4)\}$

Hence, $P(E) = P(E_3E_6) + P(E_6E_3) + P(E_4E_5) + P(E_5E_4)$

$$= P(E_3)P(E_6) + P(E_6)P(E_3) + P(E_4)P(E_5) + P(E_5)P(E_4)$$

$$= 2P(E_3)P(E_6) + 2P(E_4)P(E_5)$$

{Since E_1, E_2, E_3, E_4, E_5 and E_6 are independent}

$$= 2\left(\frac{1-k}{6}\right)\left(\frac{1+k}{6}\right) + 2\left(\frac{1+k}{6}\right)\left(\frac{1-2k}{6}\right)$$

$$= \frac{1}{18}\left[2 - k - 3k^2\right]$$

Since, $\dfrac{1}{9} \le P(E) \le \dfrac{2}{9}$

$\therefore\ -\dfrac{1}{3} \le k \le 0$

$\therefore$ Set of integral value of k = {0}

$\therefore$ Number of integral solution of k is 1